

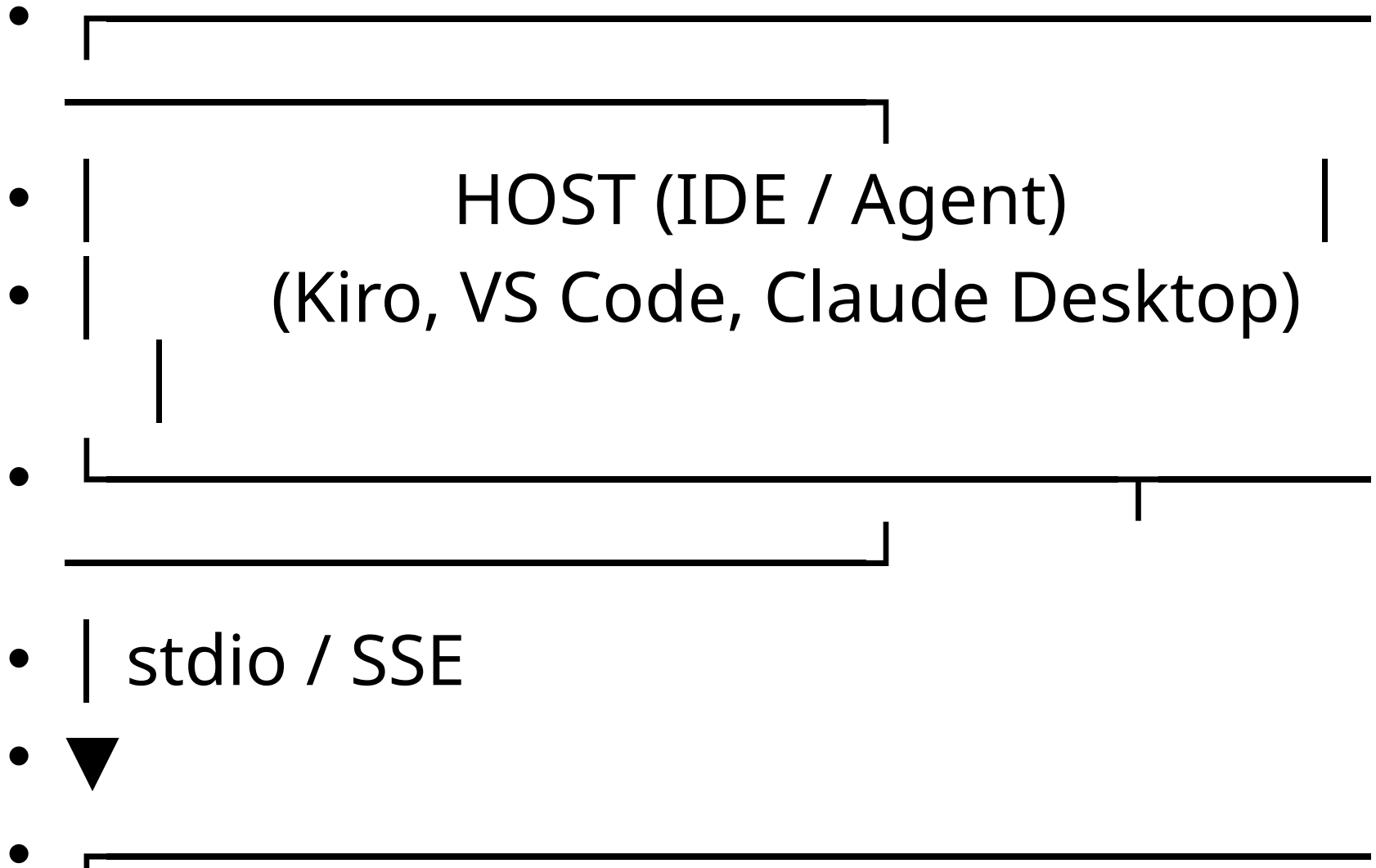
MCP Server Architecture - Block Diagram

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MCP Server Architecture

- Block Diagram Overview
- Host  MCP Server  Sub MCP Servers
- This presentation illustrates the hierarchical architecture
- of the Model Context Protocol (MCP) server system.

System Block Diagram



Sub MCP Server #1: Object Detection & Tracking

- Purpose: Computer vision-based object detection and tracking
- Capabilities:
 - Real-time object detection (YOLO, SSD, Faster R-CNN)
 - Multi-object tracking (DeepSORT, ByteTrack)
 - 3D bounding box estimation
 - Monocular depth estimation

Sub MCP Server #2: VLA-based Solution

- Purpose: Vision-Language-Action model integration
- Capabilities:
 - Visual scene understanding via VLMs
 - Natural language instruction parsing
 - Action planning and execution
 - Multi-modal reasoning (image + text → action)

Sub MCP Server #3: Local File System

- Purpose: File system access and management
- Capabilities:
 - • Read/write files and directories
 - • Search files by name or content
 - • Watch for file changes
 - • Manage project structure

Communication Flow

- Host (IDE/Agent)
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- | 1. User request (natural language)
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- MCP Server (Gateway)
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- | 2. Route to appropriate sub-server
- | 3. Tool call via JSON-RPC over stdio
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Summary

- Key Takeaways:
 - MCP provides a standardized protocol for tool integration
 - Sub MCP servers enable modular, domain-specific capabilities
 - The gateway pattern allows routing and orchestration
 - Each sub-server is independently deployable and testable